Fudan University

复旦大学

2008-2009学年第一学期考试试卷

课程名称: 离散数学(I) 课程代码: SOFT130039.01
开课院系: 软件学院 考试形式: 开卷

姓 名:_____ 学 号:_____ 专 业:_____

题目	1	2	3	4	5	6	7	8	9	10	总分
得分											

DIRECTION: There are totally **two** pages of examination paper. You must write all your answers, include your name and student number clearly, on a specific answersheet. You have 2 hours to solve all the questions.

1. Fill in the blank(For the problem with calculation, intermediate expression is enough.):(12 marks)
   a) A sequence can be constructed by flipping a coin 10 times. There are ( ) different sequences with at least for consecutive heads. (6 marks)
   b) Let $f(x)$ be the generating function of the recurrence relation $a_n = 2a_{n-1} + 3a_{n-2} - a_{n-3}$, where $n \geq 3$ and $a_0 = 2, a_1 = -1, a_2 = 3$. Then the generating function $f(x) =$( ).(6 marks)

   Solution. a) First consider a sequence with exactly four consecutive heads. We can apply inclusion-exclusion principle. Then the total number is $7 \times 2^6 - 6 \times 2^5 + 5 \times 2^4 - 4 \times 2^3 + 3 \times 2^2 - 2 \times 2 + 1$.
   b) The generating function is $f(x) = \dfrac{2 - 5x - x^2}{1 + 2x - 3x^2 + x^3}$.

   □

2. Let H be a subgroup of a group G. Prove that the relation defined by the rule aRb if $b^{-1}a \in H$ is an equivalence relation on G. (10 marks)

   Solution. H is a subgroup of G, it means $e \in H$(1 mark). Now to verify:
   a) Reflexive: $a^{-1}a = e \in H$(1 mark).
   b) Symmetric: aRb means $b^{-1}a$. While bRa means $a^{-1}b = (b^{-1}a)^{-1}$. $b^{-1}a \in H$ means $a^{-1}b \in H$, for H is a group.(1 mark)

c) Transitive: If aRb, bRc, we have $b^{-1}a, c^{-1}b \in H$. And $c^{-1}bb^{-1}a = c^{-1}a \in H$, which means aRc.(1 mark)

□

3. Given two graph G_1, G_2, prove that $\max\{\chi(G_1), \chi(G_2)\} \leq \chi(G_1 \cup G_2) \leq \chi(G_1) \cdot \chi(G_2)$, where $\chi(G)$ means vertex coloring of G. (10 marks)

Solution. We first prove the left inequality. If G_1 and G_2 have no common vertices. It is obvious that G_1's colouring will not affact G_2's coloring after merge of them. Then the new graph can be coloured by $\max\{\chi(G_1), \chi(G_2)\}$. (2 marks)

Suppose that there are common vertices of G_1, G_2. We just consider the vertices have the same color in G_1, which means that they are not adjacent each other in G_1. However if G_2 is merged, it may make some vertices adjacent. Inasmuch as G_2 can be colored by $\chi(G_2)$ colors, we can color these vertices in the same color with $\chi(G_2)$ different colors. And there are totally $\chi(G_1)$ colors in G_1. The right inequality holds also. (3 marks) □

4. Show that it is impossible, using 1×2 rectangles (dominoes), to tile an 8×8 square (chessboard) from which two opposite 1×1 corner squares have been removed.(Hint:*use maximum matching.*) (10 marks)

Solution. We can put two different colors, say black and white, on each 1×1 of a domino. If the modified square can be fully covered, there are equal number of black and white.(1 marks) We can prove that any feasible cover is isomorphic to this cover, where any two adjacent 1×1 have different color or except two. You can image that the direction of domino could be changed to avoid collision of color one by one.(1 marks)

Finally, the two remained corners must have the same color.(1 mark) And the remainders have a full cover.(1 mark) It means that one part has two more 1×1 than another part, which is a maximum matching.(1 mark) The proof is finished. □

5. G is a connected simple graph. (12 marks)
   a) G has a point whose removal does not disconnect the graph.
   b) If G is directed and strongly connected graph, prove or disprove the same question.

Solution. a) Let $P = (x_0, x_1, \ldots, x_m)$ be a maximum path in G. Suppose $G - x_0$ is not connected. Let G_1 be component of $G - x_0$ which does not contain $P - x_0$. Since G is connected, there must be an edge (y, x_0) joining G_1 to x_0. Now $(y, x_0, \ldots, x_m)$ is a longer path than P.

b) A cycle show that the assertion does not hold for strongly connected digraphs.

□

6. Prove or disprove the following statements. If it is right, you should give a proof otherwise a counterexample:(12 marks)
   a) An integral domain with a finite number of elements is a field. (7 marks)
   b) Every connected graph contains a walk that traverse each of its edges exactly once in each direction. Here, a *walk* in a graph G is a non-empty alternating sequence $v_0e_0v_1e_1 \dots e_{k-1}v_k$ of vertices and edges in G such that $e_i = \{v_i, v_{i+1}\}$.(5 marks)

   Solution. a) Let $D = \{x_0, x_1, x_2, \dots, x_n\}$ be a finite integral domain with x_0 as 0 and x_1 as 1. We have to show that every nonzero element of D has a multiplicative inverse. If x_i is nonzero, we show that the set $x_iD = \{x_ix_0, x_ix_1, x_ix_2, \dots, x_ix_n\}$ is the same as the set D. If $x_ix_j = x_ix_k$, then, by the cancellation property, $x_j = x_k$. Hence all the elements $x_ix_0, x_ix_1, x_ix_2, \dots, x_ix_n$ are distinct, and x_iD is a subset of D with the same number of elements. Therefore, $xiD = D$. But then there is some element, x_j, such that $x_ix_j = x_1 = 1$. Hence $x_j = x_i^{-1}$, and D is a field.
   b) Because if each edge is visited twice, there must be a Euler tour. Then the result is obviously true.

□

7. Given a group G and two subgroup K and H, prove the following statements: (12 marks)
   a) The intersection $K \cap H$ is a subgroup of H. (4 marks)
   b) If K is a normal subgroup of G, then $K \cap H$ is a normal subgroup of H. (8 marks)

   Solution. a) First, $e \in K \cap H$.(1) Second, if $a \in K \cap H$, a^{-1} also in $K \cap H$.(1) So it is group.
   b) As we know $L = K \cap H$ is a subgroup.(1) Given $l \in L$ and $h \in H$, consider $a = hlh^{-1}$. Because $l \in K$, which is a normal group, then $a \in K$.(1) H is a group and $l \in H$, then a must be in H.(1) It means $a \in L = K \cap H$.(1) So $K \cap H$ is a normal subgroup.

□

8. Define homomorphism of fields, and prove that every homomorphism of fields is one-to-one(injective). (10 marks)

   Solution. a) Given two fields F_1, F_2, a function $f : F_1 \to F_2$ is a homomorphism if it satisfy the following rules: 1) $f(a + b) = f(a) + f(b)$, (1)2) $f(a \times b) = f(a) \times f(b)$.(1)

b) Suppose $a \neq b$ and $f(a) = f(b)$, we have $f(a-b) = f(a)-f(b) = f(0)$. Because $a - b \neq 0$, we have $f(1) = f((a-b)(a-b)^{-1} = f(a-b)f((a-b)^{-1}) = f(0)$.(1) But $f(1) = f(0)$ is impossible.(1) So for any $a \neq b$, $f(a) \neq f(b)$, which means f is injective.(1)

□

9. R is the set of continuous functions $\mathcal{R} \to \mathcal{R}$. Addition and multiplication are defined as following

   - $[f + g](x) = f(x) + g(x)$,
   - $[f \circ g](x) = f(g(x))$.

   Determine whether the given structure forms a ring. If it is not a ring, determine which of the ring requirements hold and which fail. (12 marks)

   Solution. It is a ring(1 mark). Easy to verify the association rule of $+$ and $\circ$(2 marks). The inverse of $(f, +)$ is $-f$(1 mark). The identity of $(f, \circ)$ is $I(x) = x$(1 mark). And the distributive law is also easy to verify(1 mark). □